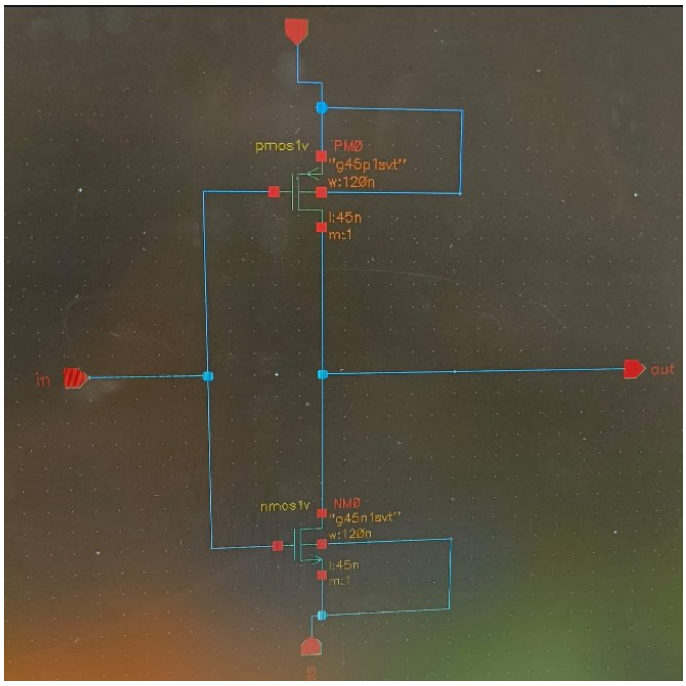
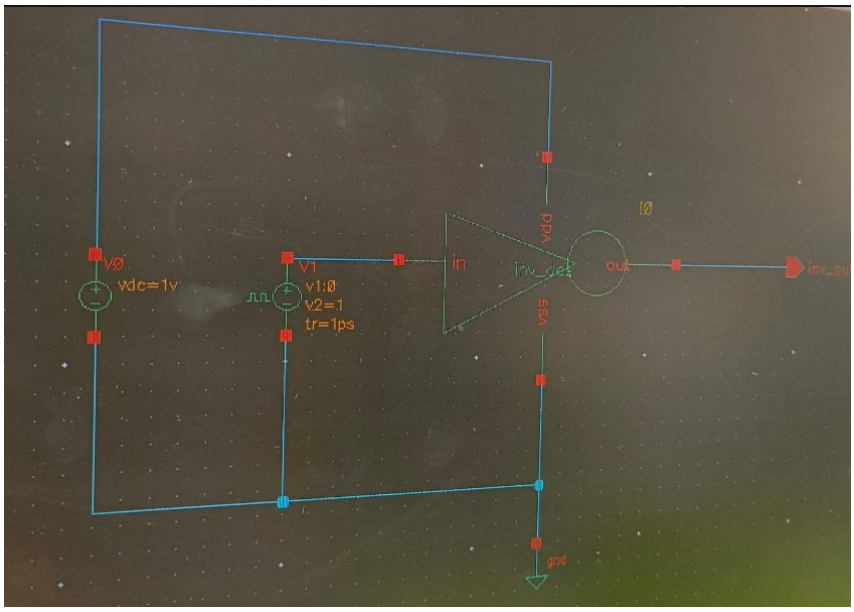
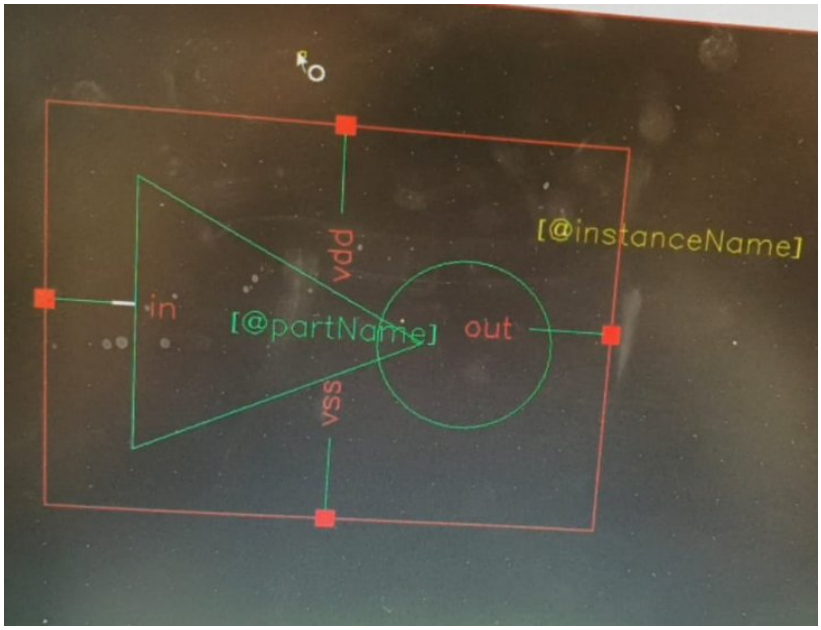
EXPERIMENT 1

Simulation-Based Analysis of N-Well CMOS Fabrication Techniques Using Cadence Design Environment

Circuit diagram:



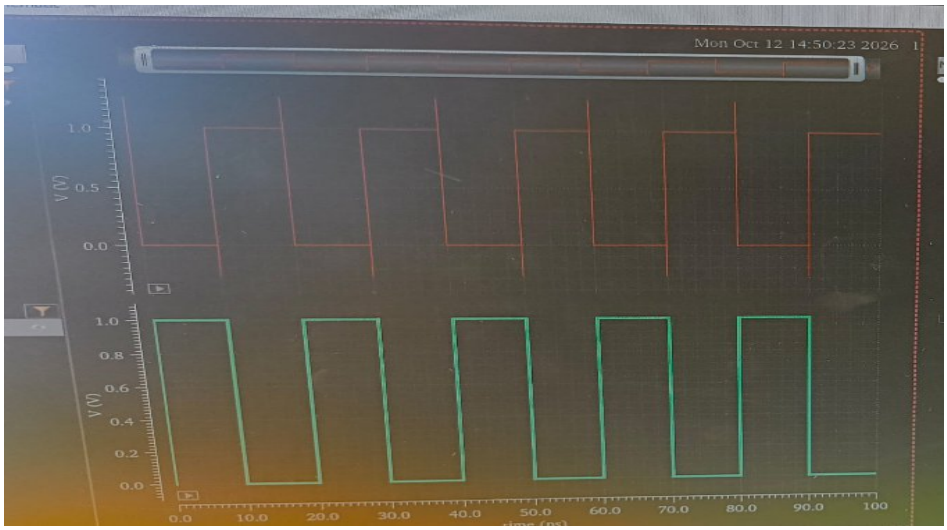




Output Parameter Table:

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Fabrication sequence	Standard CMOS process flow	Standard CMOS fabrication sequence observed	0% deviation
2	Gate length	180 nm	180 nm	0% deviation
3	Gate oxide thickness	4 nm	4 nm	0% deviation
4	Source/Drain formation	Properly defined regions	Source and drain regions properly formed	0% deviation
5	Layer structure	Correct CMOS layers	Substrate, gate oxide, gate, source/drain and metal interconnect layers identified	0% deviation
6	Device dimensions	As per design	Record simulated device dimensions from Cadence	0% deviation

Output:


Mon Oct 12 14:50:23 2026
V (V)
1.0
0.5
0.0
V (V)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100